

The Grammar of some Rootless Structures

Neil Myler
myler@bu.edu
Boston University

Roots IV, New York University
July 1 2015

Introduction

- **Teaser:** Possession sentences exhibit *too many (surface) structures* for one meaning (Heine 1997; Stassen 2009).

- (1) I have a book.
 - (2) Ég er með bók. [Copular be verb + possessee in with PP]
I am with book.ACC
'I have a book (lit: 'I am with a book').' (Icelandic- see Levinson 2011)
 - (3) u menja est' kniga. [Existential be + locative PP possessor]
At me.GEN is book
'I have a book (lit: 'at me is a book').' (Russian)
 - (4) Nekem van könyvem. [Existential be + Possessive DP]
I.DAT is book.1POSS.NOM
'I have a book (lit: 'there is a book of mine').' (Hungarian- see Szabolcsi 1981)
 - (5) Noqa-qta libro-(y) tiya-pu-wa-n. [Existential be + Appl]
I-GEN book-1POSS be_{exist}-APPL-1OBJ-3S
'I have a book (lit: 'there is a book for me').' (Cochabamba Quechua)
- A stern test for any theory of how thematic roles relate to structural positions, especially things like...
- (6) The UNIFORMITY OF THETA ASSIGNMENT HYPOTHESIS (UTAH- Baker 1988:46)
Identical thematic relationships between items are represented by identical structural relationships between those items at the level of D-structure.

Two broad options:

- **One option:** Assume the possessor role is assigned in the same place in all languages (perhaps inside the DP, like in Hungarian), and derive (1) by movement. (Freeze 1992, Kayne 1993; many others).
 - This preserves UTAH.
 - **Interesting Prediction:** Despite looking transitive, HAVE is secretly unaccusative.
- **Second option:** Assume that the possessor is being introduced in different places in different languages: in spec-VoiceP in the case of HAVE (Beavers, Ponvert and Wechsler 2008; Belvin 1996; Myler 2014; Partee 1999; Saebo 2009; *i.a.*).
 - Entails abandoning UTAH; the regularities captured by UTAH must instead be captured by the interaction of syntactic structure with compositional semantics.
 - **Different Prediction:** HAVE is what it looks like– a transitive verb.
- **This talk:** The second broad option is the right one, because its prediction about HAVE turns out to be correct.

Roadmap

1. Two big questions in argument structure
2. Thematic roles out of syntax: Deriving (some of) the interpretations of HAVE
3. HAVE is not secretly unaccusative
4. Conclusion: Structural meaning in rootless structures

1 Two big questions in argument structure

- (7) What are thematic roles? Are they syntactic features that need to be “checked”/“valued” (Hornstein 1999 et seq.), or are they just (parts of) the meanings of syntactic heads (Heim & Kratzer 1998, i.a.)?
- (8) What aspects of meaning come from roots, and what aspects come from the structure roots are embedded in?

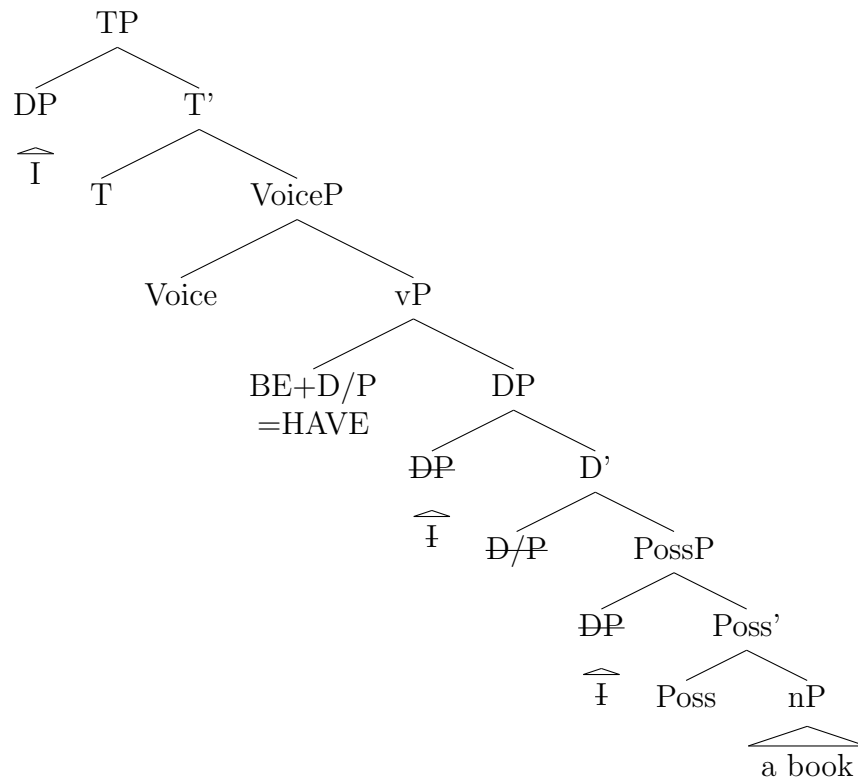
Answers from this talk:

- The properties of possession sentences seem to indicate that thematic roles are purely semantic; they are not syntactic features that need to be checked in particular positions.
- One way of attacking question (8) is to find structures which have no lexical root at all, and examine their properties.
 - As many have pointed out before, light verbs are the place to look, and HAVE and BE might be the ideal test case (since they are arguably the lightest verbs of all; see Belvin 1996; Ritter and Rosen 1993, 1997; Harley 1995, 1998, 2002; *i.a.*)
 - I’ll argue that HAVE sentences add support to a view of the syntactico-semantic contribution of Voice which others have reached on independent grounds: namely that Voice’s semantic contribution is determined by its surrounding structural context.
 - In particular, Voice can be expletive, or it can introduce an Agent or Holder role depending on the denotation of its complement (Alexiadou, Anagnostopoulou and Schaefer 2015, Kratzer 1996; Schaefer 2008; Wood 2015a,b)

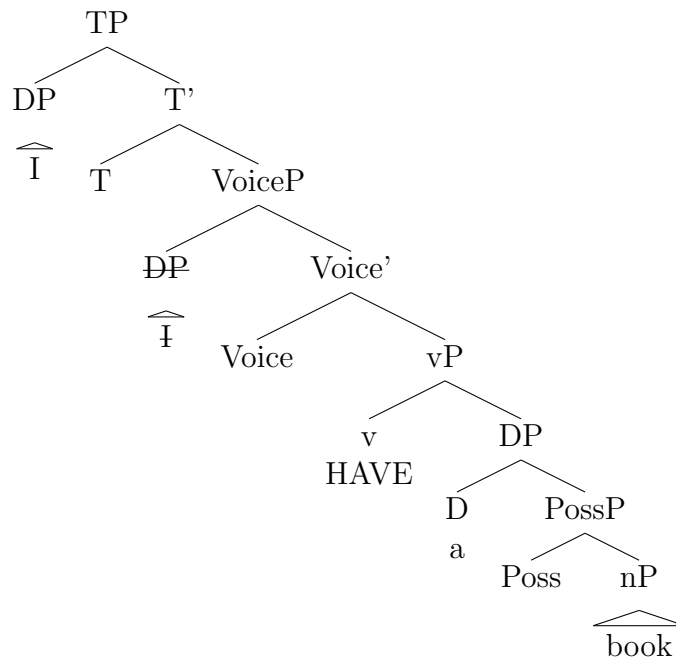
2 Thematic roles out of syntax: Deriving (some of) the interpretations of HAVE

Reminder: Two broad options for dealing with the fact that transitive HAVE expresses the same thematic relations as something which Hungarian does by putting a possessed DP underneath BE:

- (9) UTAH-Preserving Solution (cf. Freeze 1992; Kayne 1993)



- (10) Alternative Solution (today)



Rest of this section:

- I will give the alternative solution in (10) legs by showing how it derives some of the various interpretations of HAVE.
- See Myler (2014) for derivations of other interpretations which we don't have time for here, many of which have antecedents in Beavers, Ponvert and Wechsler (2008); Belvin (1996); Partee (1999); Saebo (2009).
- Having shown that (10) works, in section 3 I will demonstrate that it is superior to (9) with respect to the predictions it makes about the argument structure of HAVE.

2.1 Key Assumptions

(11) Assumptions about Possession Relations and the HAVE/BE Relation

- a. Possession is fundamentally a relationship between two DPs- the possessor and the possessee. (Szabolcsi 1981, 1994; Kayne 1993; Partee 1999).
- b. To link such a relation to tense, clause type, etc., you need a copula- a dummy verb which simply denotes a type-neutral identity function.
- c. Copulas exist to “sentencify” fundamentally non-sentency meanings (Pustet 2003, Tham 2013).
- d. The copula is realized as HAVE if the rest of the structure is transitive (Hoekstra 1994); it will be realized as BE otherwise.

(12) General Assumptions about Functional Heads

- a. A given functional head may {require/forbid/optionally allow} a specifier, as a matter of (micro-)parametric variation.
- b. Thematic roles are not syntactic features that need to be checked, they are just part of the meanings of heads. Hence, the presence/absence of a specifier is independent of whether a head introduces a thematic role.

(13) The Allosemy of Voice (Adapted from Wood 2015a:30, see also AAS 2015; Schaefer 2008; Wood 2015b)

- a. $\llbracket \text{Voice} \rrbracket \Leftrightarrow \lambda x_e. \lambda e_s. \text{Agent}(x, e) / __\text{(agentive, dynamic event)}$
- b. $\llbracket \text{Voice} \rrbracket \Leftrightarrow \lambda x_e. \lambda e_s. \text{Holder}(x, e) / __\text{(stative eventuality)}$
- c. $\llbracket \text{Voice} \rrbracket \Leftrightarrow \lambda x. x / __\text{(elsewhere)}$

- This collection of assumptions, all of which have been motivated independently of the proposal in (10), yield the following predictions about the possible interpretations of HAVE sentences.

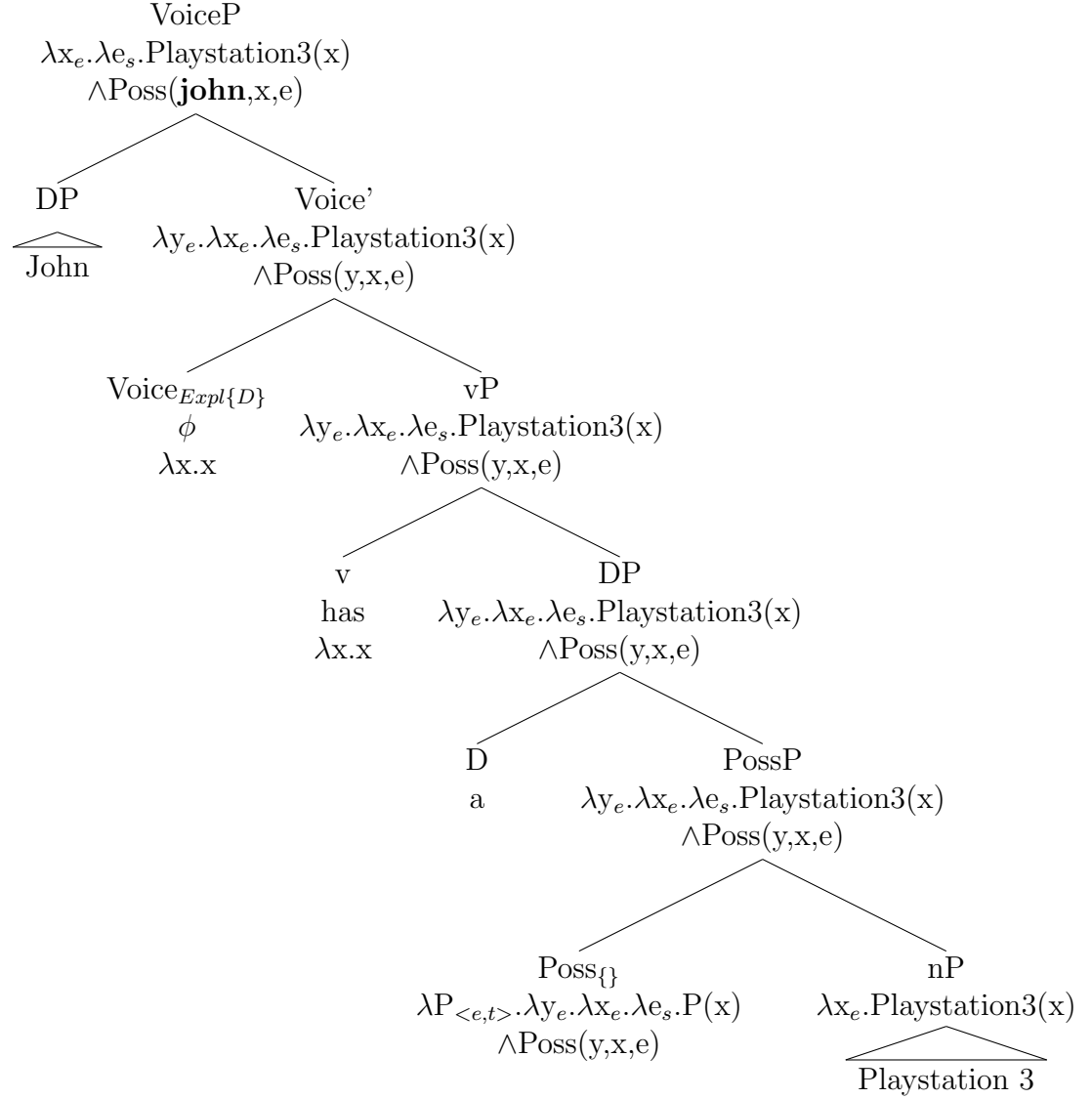
(14) Predicted Interpretations of HAVE constructions

- a. Cases where the meaning of a *have* sentence = that of *have*'s complement (if Voice = Expl)
- b. Cases where the meaning of a *have* sentence = that of *have*'s complement+that of Voice (if Voice $\neq$ Expl)

2.2 Cases where the meaning of a *have* sentence = that of *have*'s complement (if Voice = Expl)

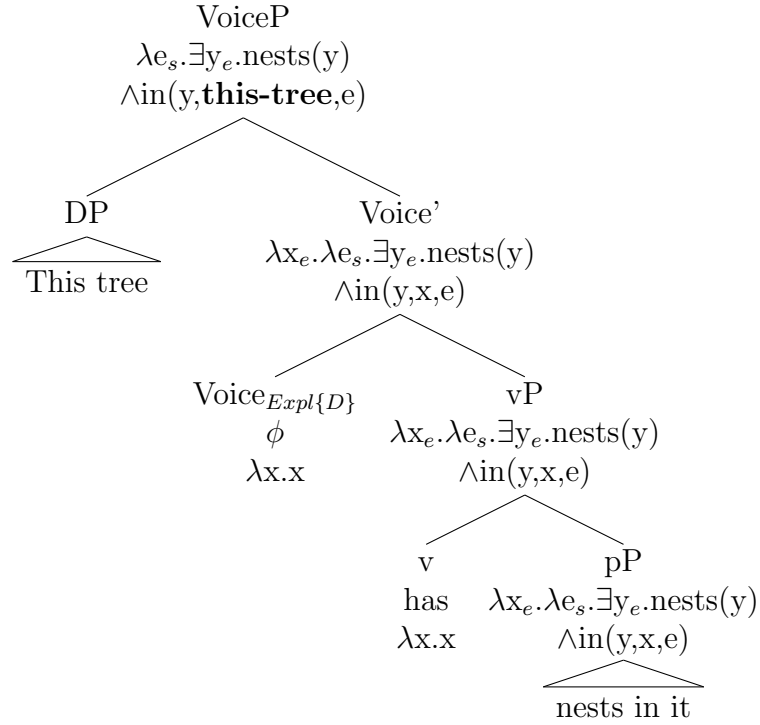
- Cases of this sort include (i) the HAVEs of permanent ownership, as well as inalienable HAVE; (ii) so-called locative HAVE.

(15) John has a Playstation 3.



- The open individual variable corresponding to the possessee subsequently undergoes existential closure (Diesing 1992).
- Note that the possessor thematic role is introduced by Poss, but not saturated until spec-VoiceP. We might call this *delayed gratification*.
- An imaginable alternative (anticipated in Ritter & Rosen 1997:313) would be to have Poss introduce a silent pronominal bound by the possessor in spec-VoiceP. This makes an incorrect prediction about HAVE languages which have DP-internal possessor agreement (see appendix).

(16) This tree has nests *(in it).



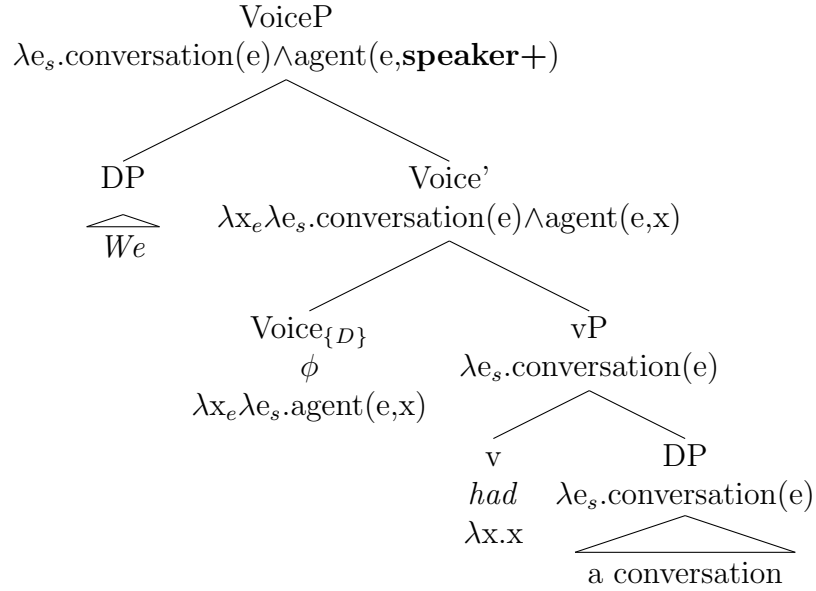
2.3 Cases where the meaning of a *have* sentence = that of *have*'s complement+that of Voice (if Voice $\neq$ Expl)

- Cases of this sort include (i) HAVE+simplex event nominals and (ii) stative causative HAVE.

(17) LIGHT VERB *have*

- We had a conversation.
- John had an accident.
- We had an argument.
- I had a bath.
- We're having fun.
- You'll have trouble getting that to work.

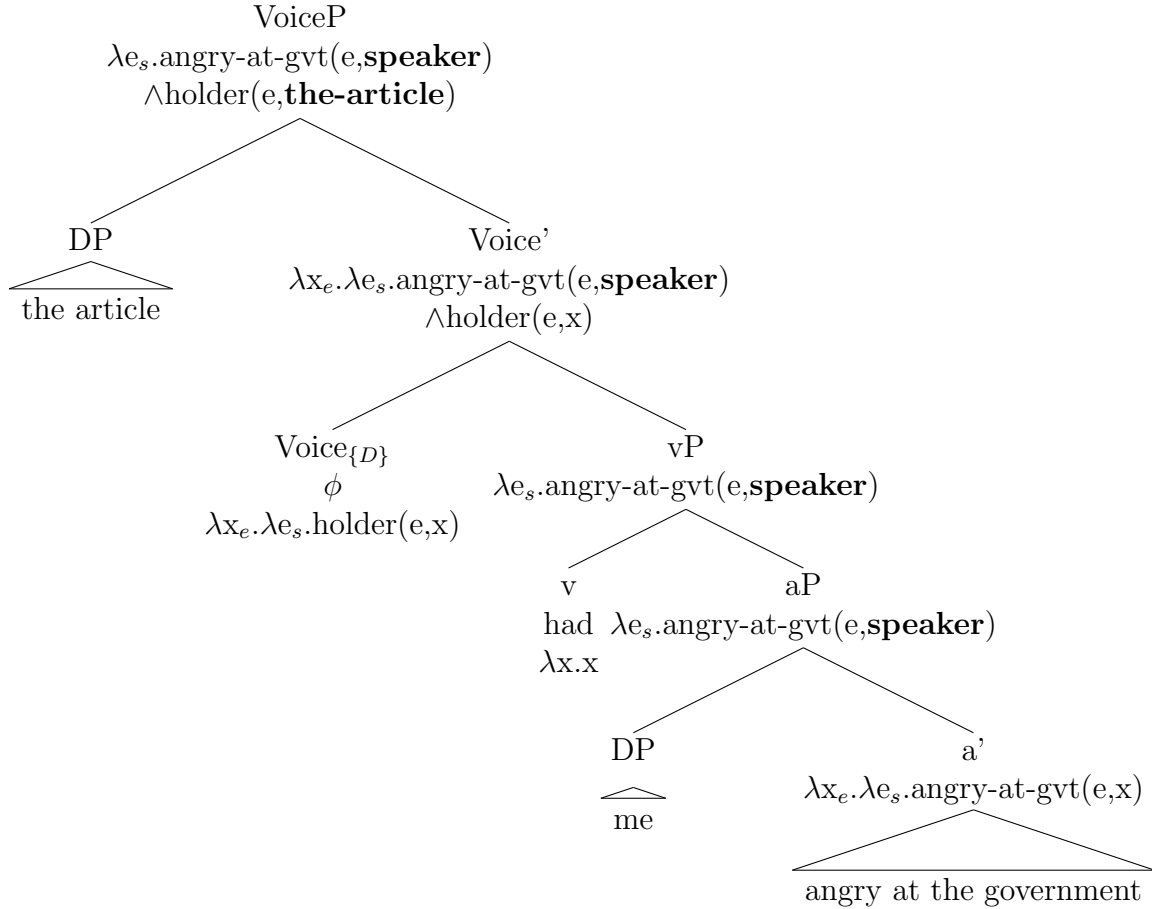
(18) We had a conversation.



• Causer HAVE:

- (19) a. The article had me angry at the government/weeping in fury.
 b. Superman has the bad guy pinned to the floor/begging for mercy.

(20) The article had me angry at the government. (Causer *have*)



- Here I assume that “causer” can be a special case of the *holder* role, as is also found in stative readings of verbs like *worry*.
- Additional payoff: we correctly predict that HAVE sentences are stative or eventive depending on whether the complement of HAVE is stative or eventive (Belvin 1996; Harley 1998; 2002).

2.4 Conclusion

- The alternative approach to HAVE outlined here:
 - explains how a transitive structure can come to have the same meaning as something that involves intransitive structures in other languages (via delayed gratification).
 - extends successfully to a number of other, non-possessive uses of HAVE.
- Now that we know this approach is workable, let’s go on to an argument that it is better than the standard approach.

3 HAVE is not secretly unaccusative

- The UTAH-preserving approach to HAVE predicts that HAVE sentences are hidden unaccusatives.
- The alternative outlined in the previous section predicts that HAVE sentences are true transitives.
- I will now show that...
 - the traditional arguments in favor of HAVE being unaccusative are not demonstrative
 - there are a number of arguments that point in the opposite direction.

3.1 HAVE doesn’t passivize (?)

- As is well known, possessive uses of HAVE do not passivize. This looks like evidence that HAVE is indeed unaccusative.

(21) * {a sister/a Playstation 3/red hair} was had by John.

- It turns out that the possibility of passive with at least English *have* is more widespread than many suppose.
- Moreover, the cases where passivization work have a systematicity to them which precludes writing them all off as idioms.
- Specifically, passivization of *have* is often acceptable so long as the complement of *have* is event-denoting.

(22) **Light verb *have* can be passivized**

- a. A terrible fight was had at that street corner.
- b. A thorough discussion needs to be had before we proceed.
- c. A debate was had to resolve the issue.
- d. He's unlikely to leave while there's still fun to be had.

- **Proposal:** the failure of passive with possessive HAVE is actually a semantic problem, arising from the denotation of the passive morpheme.
- Bruening (2013): passive takes a function of type $\langle e\langle s,t \rangle \rangle$ as its first argument.
 - Voice' will have the right sort of denotation in cases like (22), because the complement of HAVE itself denotes a set of events.
 - In possessive HAVE cases, however, Voice' has the type $\langle e\langle e\langle s,t \rangle \rangle \rangle$ (recall (15)), producing a fatal type mismatch which rules out (21).
- Open question: why can't ECM variants of HAVE passivize?

- (23) a. Mary had John leave.
b. * John was had (to) leave by Mary.

3.2 (Non-modal) HAVE as similar to a raising verb (?)

- A general expectation of the standard, UTAH-preserving approach to HAVE is that it is essentially a raising verb which takes a DP rather than a TP complement.
- In other words, possessive HAVE should pattern with SEEM etc. in interesting respects.
- **Prediction:** DP-sized idioms which include a possessor should have counterparts in HAVE.
- Such idioms are not numerous in English, and the ones that exist have some special properties which make it hard to conclude anything from them, but it is at least suspicious that this prediction is not borne out.

- (24) a. the cat's pyjamas. = 'the feline's sleepwear/something outstanding'
b. the pyjamas that the cat has = 'the feline's sleepwear/*something outstanding'
c. The cat has pyjamas. = 'the feline has sleepwear/*this is outstanding'
- (25) a. the bee's knees. = 'the bit of *Apis mellifera*'s body corresponding to our *articulatio genus*/something outstanding'
b. the knees that the bee has. = 'the bit of *Apis mellifera*'s body corresponding to our *articulatio genus*/*something outstanding'
c. The bee has knees. = *Apis mellifera*'s body has parts corresponding to our *articulatio genus*/*this is outstanding'

- (26) a. the dog's bollocks. = 'the canine's testicles/something outstanding'
 b. the bollocks that the dog has = 'the canines testicles/*something outstanding'
 c. The dog has bollocks. = 'the canine has testicles/*this is outstanding'
- (27) a. the devil's advocate. = 'Satan's lawyer/the person deliberately arguing a contrary position'
 b. the advocate that the devil has = 'Satan's lawyer/*the person deliberately arguing a contrary position'
 c. The devil has an advocate. = 'Satan has a lawyer/??there is a person deliberately arguing a contrary position'

- Another problem comes from Imbabura Quechua, where embedded clauses can take structural case-marking (since they are nominalized):

- (28) IMBABURA QUECHUA- TRANSITIVE CLAUSE-TAKING VERB (Jake 1985:158, her (63a))

Maria-ka cri-n [Francisco kaypi ka-j]-ta.
 Maria-TOP believe-3S Francisco here be-NOM-ACC
 'Maria believes that Francisco is here.'

- The SEEM verb in Imbabura Quechua, as one might expect, does not assign accusative case in this way.

- (29) IMBABURA QUECHUA- SEEM DOES NOT ASSIGN ACC (Hermon 2001:160, her (38))

kan_i-ka [t_i puñu-naya-y] yari-ø-ngi.
 you-TOP t_{you} sleep-IMPULS-INF seem-PRES-2S
 'You seem to want to sleep.'

- Importantly, Imbabura Quechua is a HAVE language. And its HAVE verb does not pattern with SEEM with respect to case assignment.
- Instead, it assigns accusative.

- (30) IMBABURA QUECHUA- HAVE DOES ASSIGN ACC (Cole 1982:94, his (373))

Juzi iskay kaballu-ta chari-ø-n.
 Jose two horse-ACC have-PRES-3S
 'Jose has two horses.'

- Again, this is unexpected on the standard, UTAH-preserving account of HAVE.

3.3 The unaccusative analysis predicts that *haveable* should be unhaveable

- Oltra-Massuet (2010), Wood & Sigurðsson (2014): *-able* can be productively used only with transitive verbs, and not unaccusative ones.¹
- If so, then examples like the following (retrieved from the web on December 13th 2013) would seem to indicate that the unaccusative analysis of HAVE makes the wrong prediction.

(31) ...the thesis can be defended that the addiction to the **haveable**, which characterizes the affluence variety of externalization, is reinforced not only by...

3.4 Auxiliary Selection

- Languages in which the perfect auxiliary varies between HAVE and BE depending on unaccusativity (such as French, Italian, German, and Dutch) always select HAVE with main verb HAVE— one would expect BE if HAVE were unaccusative.
- This is another property which differentiates HAVE from SEEM, which does take BE in Italian.

¹Richard Kayne points out to me a few apparent exceptions to this generalization. Interestingly, however, these exceptions all implicate unaccusative configurations which are independently pseudopassivizable.

- (i) That kind of mat isn't fallable on/This mat has been fallen on by many stuntmen.
- (ii) That kind of solution isn't arrivable at using these kinds of methods/This solution was arrived at only after much discussion.

If the configuration is not pseudopassivizable, it will not submit to *-able* affixation either. Note that non-motion uses of *arrive at* are different from motion uses in this respect.

- (iii) *The station isn't arrivable at by bus/*The station was arrived at by the passengers.
- (iv) *The pub isn't comable to without friends/*This pub is come to by many undergraduates.
- (v) *This bridge isn't goable over without a guide/*The bridge was gone over by many cars.

Whatever exceptional property licenses pseudopassives in configurations that allow it, I presume that it will explain the exceptions to Oltra-Massuet and Wood & Sigurðsson's claims also.

3.5 Genitive of Negation in Polish (h/t Stephanie Harves)

- Stephanie Harves points out to me that the genitive of negation in Polish is only allowed on the direct objects of transitive verbs.
- This is different from the situation in Russian, where genitive of negation is a diagnostic of unaccusativity (Harves 2002; Pesetsky 1982).
- By this diagnostic, the Polish HAVE verb is clearly transitive rather than unaccusative ((32a) is drawn from Błaszczak 2007:325, her (4.23)).

(32) POLISH GENITIVE OF NEGATION

- a. Samochód ma silnik.
car.NOM has engine.ACC
'The car has an engine.
- b. Samochód nie ma silnik-*(a).
car.NOM NEG has engine-GEN
'The car doesn't have an engine.
- c. *Samochódu nie ma silnik-(a).
car.GEN NEG has engine.GEN
'The car doesn't have an engine.

3.6 Conclusion: HAVE is not secretly unaccusative

- Contrary to the predictions of the UTAH-Preserving theory, HAVE is not secretly unaccusative after all.
- Since it was the impetus of UTAH which led to this incorrect prediction, suspicion is cast on the whole project of trying to have thematic roles "assigned" rigidly to particular structural positions.
- We thus have an argument in favor of the alternative approach in which thematic roles are simply (parts of) the meanings of syntactic heads.
- A major task, of course, is nailing down the right meanings for those heads (see next section).
- Once (if!) we manage this, the regularities in theta-role assignment captured by UTAH are then just a special case of compositionality at work.

4 Conclusion: Structural meaning in rootless structures

- (33) What are thematic roles? Are they syntactic features that need to be “checked” / “valued” (Hornstein 1999 et seq.), or are they just (parts of) the meanings of syntactic heads (Heim & Kratzer 1998, i.a.)?
- I’ve just argued, on the basis of HAVE, that the second of these alternatives is correct (see Myler 2014, forthcoming for a more extended argument taking in the broader typology of predicative possession).
- (34) What aspects of meaning come from roots, and what aspects come from the structure roots are embedded in?
- If the present analysis is on the right track, then HAVE is an excellent probe into a subpart of this question– since HAVE contributes no root-y meaning, we get a clearer view of what aspects of the meaning of the verb phrase come from places other than the root.
 - The discussion in section 2 suggests that the proposals of Alexiadou, Anagnostopoulou, and Schaefer (2015), Schaefer (2008), Wood (2015), and others as to the nature of the contributions of Voice are correct.
 - In particular, these proposals yield a wide-ranging account of the interpretations of HAVE, purely in terms of the interaction of the denotation of HAVE’s complement and the normal action of Voice itself.
 - Thanks for listening!

References

- Alexiadou, Artemis, Elena Anagnostopoulou and Florian Schaefer. (2015) *External Arguments in Transitivity Alternations: A Layering Approach*. (Oxford: OUP)
- Baker, Mark (1988) *Incorporation: A Theory of Grammatical Function Changing* (London: University of Chicago Press)
- Beavers, John, Elias Ponvert, & Stephen Wechsler (2009) Possession of a Controlled Substantive: Light ‘have’ and Other Verbs of Possession. (Ms, UT Austin)
- Belvin, Robert S. (1996) *Inside Events: The Non-Possessive Meanings of Possession Predicates and the Semantic Conceptualization of Events*. (PhD Thesis, University of Southern California)
- Błaszczak, Joanna (2007) *Phase Syntax: The Polish Genitive of Negation* (Habilitationsschrift: Universität Potsdam)
- Bruening, Benjamin (2013) *By* Phrases in Passives and Nominals (*Syntax* 16.1:1-41)
- Cole, Peter (1982) *Imbabura Quechua* (Amsterdam: North Holland Publishing Company)
- Diesing, Molly (1992) *Indefinites* (Cambridge, MA: MIT Press)
- Freeze, Ray. (1992) Existentials and other locatives (*Language* 68: 553-595)
- Harley, Heidi. (1995) *Subjects, events and licensing*. (PhD Thesis: MIT)
- Harley, Heidi (1998) You’re having me on!: aspects of *have*. (semanticsArchive/TY3M2M5O)
- Harley, Heidi. (2002) Possession and the double object construction (*Yearbook of Linguistic Variation* 2, pp. 29-68).
- Harves, Stephanie (2002) *Unaccusative Syntax in Russian* (PhD Thesis, Princeton University)
- Heim, I. & A. Kratzer (1998) *Semantics in Generative Grammar* (Oxford: Blackwell)
- Heine, Bernd (1997) *Possession* (Cambridge: Cambridge University Press)
- Hermon, Gabriella (2001) Non-Canonically-Marked A/S in Imbabura Quechua. In A.Y. Aikenvald, R.M.W. Dixon, and M. Onishi (eds.) *Non-Canonical Marking of Subjects and Objects* (pp.149-176) (Amsterdam: John Benjamins)
- Hoekstra, Teun (1994) *HAVE as BE Plus or minus*. In *Paths Towards Universal Grammar; Studies in Honor of Richard S. Kayne*, Guglielmo Cinque, Jan Koster, Jean-Yves Pollock, Luigi Rizzi & Raffaella Zanuttini (eds.) 199-215. (Washington DC: Georgetown University Press)
- Hornstein, Norbert (1999) Movement and Control. (*Linguistic Inquiry* 30:69-96)
- Jake, Janice (1985) *Grammatical Relations in Imbabura Quechua* (New York: Garland)
- Kayne, Richard S. (1993) Toward a Modular Theory of Auxiliary Selection. Reprinted in Kayne, Richard S. (2000) *Parameters and Universals* (Oxford: OUP) (pp.107-130)
- Kratzer, Angelika (1996) Severing the external argument from its verb. In Rooryck, Johan & Laurie Zaring (eds.) (1996) *Phrase Structure and the Lexicon* (Dordrecht: Kluwer)
- Levinson, Lisa (2011) Possessive WITH in Germanic: HAVE and the Role of P (*Syntax* 14.4:355-393)
- Myler, Neil (2014) *Building and Interpreting Possession Sentences*. (PhD Thesis: NYU)
- Myler, Neil (In Preparation) *Building and Interpreting Possession Sentences*. (Book Ms, Boston University)
- Oltra-Massuet, Isabel. (2010) *On the Morphology of Complex Adjectives* (PhD Thesis, Universitat Autònoma de

Barcelona)

Partee, Barbara H. (1999) Weak NPs in HAVE Sentences. In J. Gerbrandy, M. Marx, M. de Rijke, and Y. Venema (eds.) *JFAK, a Liber Amicorum for Johan van Benthem on the occasion of his 50th Birthday*, 39-57 (University of Amsterdam, Amsterdam)

Pesetsky, David. (1982) *Paths and Categories*. (Ph.D. Thesis: MIT.)

Pustet, Regina. (2003) *Copulas: Universals in the categorization of the lexicon*. (Oxford: Oxford University Press)

Ritter, Elizabeth & Sara Thomas Rosen (1993) Deriving causation. (*Natural Language and Linguistic Theory* 11, 519-555)

Ritter, Elizabeth & Sara Thomas Rosen (1997) The function of *have* (*Lingua* 101:295-321)

Sæbø Kjell (2009) Possession and Pertinence: the meaning of *have*. (*Natural Language Semantics* 17:369-397)

Schäfer, Florian. (2008) *The Syntax of (Anti-)Causatives*. (Philadelphia: John Benjamins)

Stassen, Leon. (2009) *Predicative Possession*. (New York, NY: Oxford University Press)

Szabolcsi, Anna (1981) The Possessive Construction in Hungarian: A Configurational Category in a Non-Configurational Language. (*Acta Linguistica Academiae Scientiarum Hungaricae*, 31, 261-289.)

Szabolcsi, Anna. (1994) The noun phrase. In *The Syntactic Structure of Hungarian*. *Syntax and Semantics* 27, ed. by Ferebe Kiefer and Katalin E. Kiss, 179-274. (San Diego, CA: Academic Press.)

Tham, Shiao Wei. (2013) Possession as non-verbal predication. (Handout from a talk given at the 39th Berkeley Linguistics Society Conference, Feb 16-17, 2013)

Wood, Jim. (2015a) *Icelandic Morphosyntax and Argument Structure*. (Dordrecht: Springer)

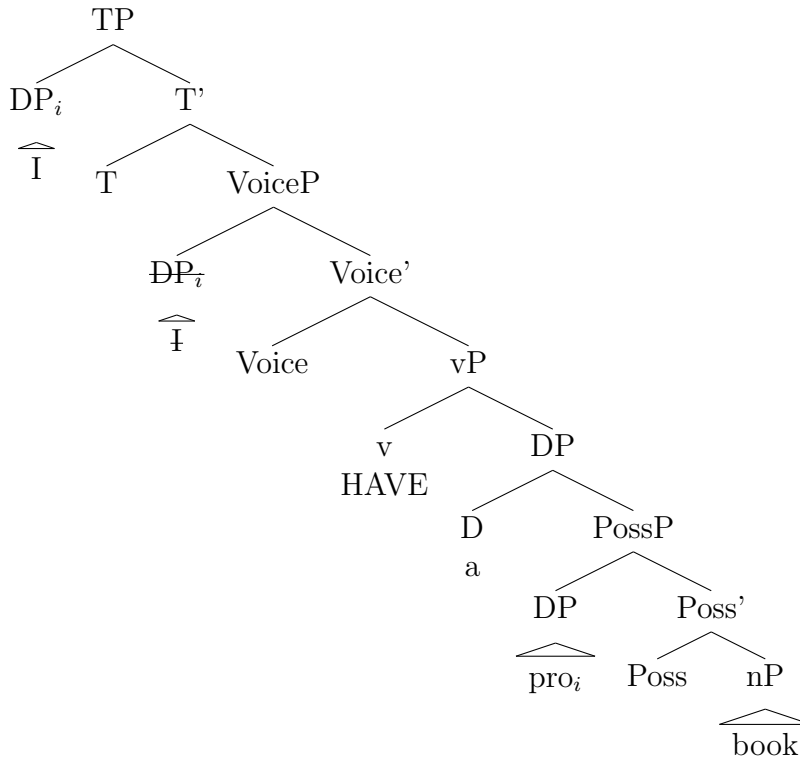
Wood, Jim. (2015b) How roots do and don't constrain the interpretation of Voice. (Talk Given at Roots IV, NYU, June 30 2015)

Wood, Jim and Einar Freyr Sigurðsson. (2014) Building deverbal ability adjectives in Icelandic. (*University of Pennsylvania Working Papers in Linguistics* 20 (1), 351-360.)

Appendix: Against a Binding-based amendment to (15)

- Ritter & Rosen (1997:313) give an approach to possessive HAVE very similar to mine, but which mediates the relationship between HAVE's subject and spec-PossP via binding, rather than delayed gratification.

(35) Ritter & Rosen-inspired Alternative to Today's Approach



- **Predictive difference:** Suppose we find a language which exhibits transitive HAVE, and also requires Hungarian-style agreement between the head noun and its possessor in possessed DPs.
- The approach in (35) predicts that HAVE sentences should require DP-internal possessive agreement, triggered by the presence of the silent pronoun.
- The delayed gratification approach makes no such prediction.
- Santiago del Estero Quechua, a HAVE language of Argentina, has the properties required to test these predictions. The facts favor the delayed gratification approach.

(36) Juan-pa pana-*(n)
 Juan-GEN sister-3SG
 'Juan's sister'

(37) Juan pana-ta api-n.
 Juan sister-ACC have-3SG
 'Juan has a sister.'